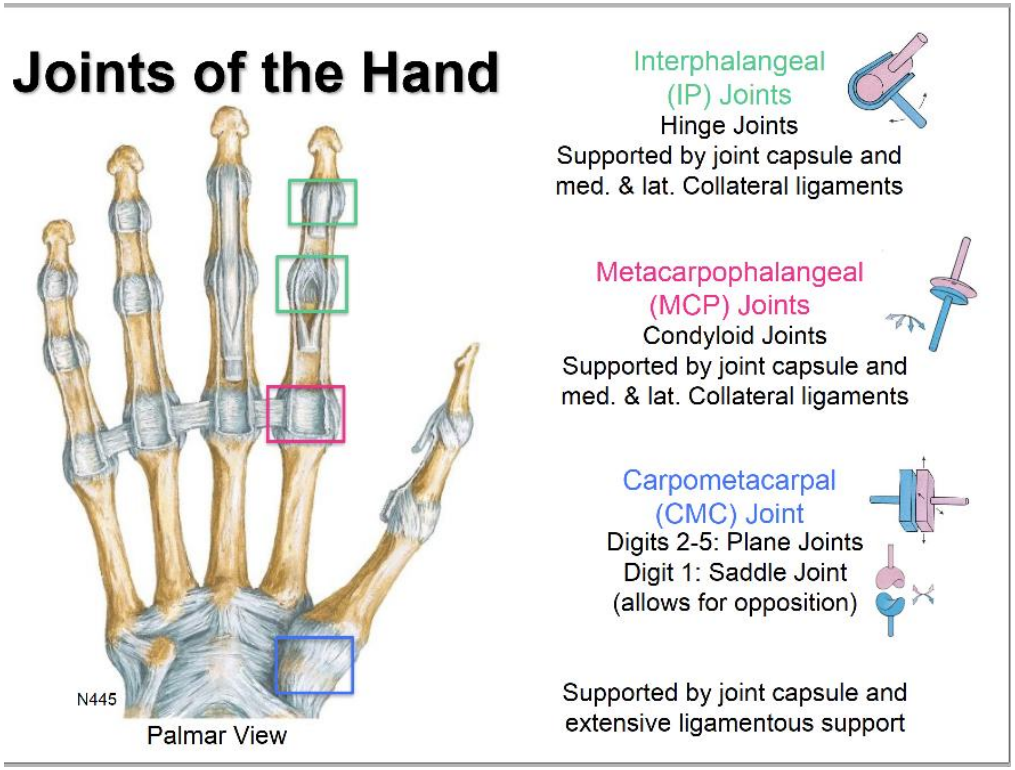
Week #9: [Wrist & Hands]

Joints of the Hand
Specific Fractures of the Wrist & Hand
- Distal Radius Fractures **(Dr. Faber)**
- Scaphoid Fractures **(Dr. Faber)**

Anatomy of the Soft Tissues of the Hand
Nerve Supply of the Hand
Carpal Tunnel Syndrome

Wrist & Hand
- Joints of the Hand



I. Hand

A. Fascial Structures of the Hand

a. Palmar Carpal Ligament
   i. Anterior thickening of antebrachial fascia, continuous with the extensor retinaculum.
   ii. More superficial & proximal to the flexor retinaculum.

Wrist & Hand: Specific Fractures of the Wrist & Hand
- Distal Radius Fractures (Dr. Faber)
- Scaphoid Fractures (Dr. Faber)

Anatomy of the Soft Tissues of the Hand

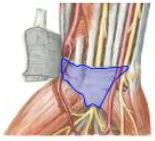
Fascial Structures



Palmar Aponeurosis

Insertion point of Palmaris Longus Tendon
Thick longitudinal fibres course from retinaculum to medial four digits

Palmar Carpal Ligament



Thickening of antebrachial fascia
Continuous with extensor retinaculum

Flexor Retinaculum

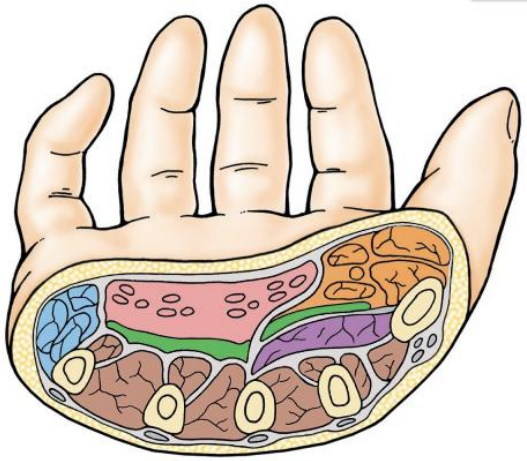
Deep to palmar carpal ligament
Roof to carpal tunnel
Extends from scaphoid & trapezium to pisiform & hook of hamate

2. Palmar Aponeurosis

a. Thick, longitudinally oriented fibers course from the distal edge of the flexor retinaculum to the medial 4 digits, becoming continuous with the fibrous tendon sheaths of the digits.
b. The palmaris longus tendon inserts into the proximal portion of the aponeurosis.

Hand Organization

Hypothenar Compartment | **Central Compartment** | **Thenar Compartment**

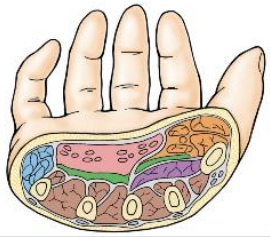


Adductor & Interosseous Compartment

Thenar Compartment



- Act upon the thumb
- Three muscles
- Make up thenar eminence
- Innervated by: Recurrent Branch of Median Nerve
  - **Exception:** 1 muscle innervated by **Deep Branch of Ulnar Nerve**

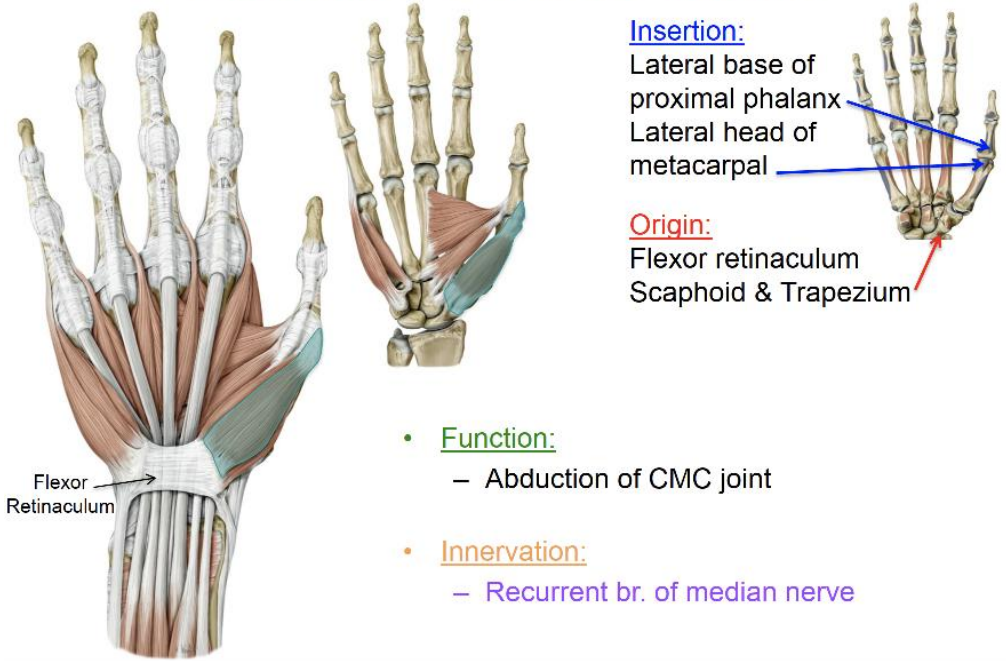


Palmar Aponeurosis

a. Thick, longitudinally oriented fibers course from the distal edge of the flexor retinaculum to the medial 4 digits, becoming continuous with the fibrous tendon sheaths of the digits.
b. The palmaris longus tendon inserts into the proximal portion of the aponeurosis.
c. Extensions (septa) from the palmar aponeurosis divide the hand into thenar (thumb), hypothenar (little finger), & central compartments.
    i. **Thenar compartment:** associated with muscles of thumb
    ii. **Hypothenar compartment:** associated with muscles of the little finger (digit 5)
    iii. **Central compartment:** deep to palmar aponeurosis; associated with forearm flexor tendons
    iv. **Adductor/Interosseous compartment:** Deepest layer

Clinical Note: Dupuytren's Contracture is an unexplained progressive shortening (contracture), thickening, and fibrosis of the collagen fibers in the palmar aponeurosis. Fingers are drawn into permanent flexion (a non-functional curled position). This fascial disease (nodules and thickened fibers) must be surgically excised so a patient can open up the palm.

Abductor Pollicis Brevis

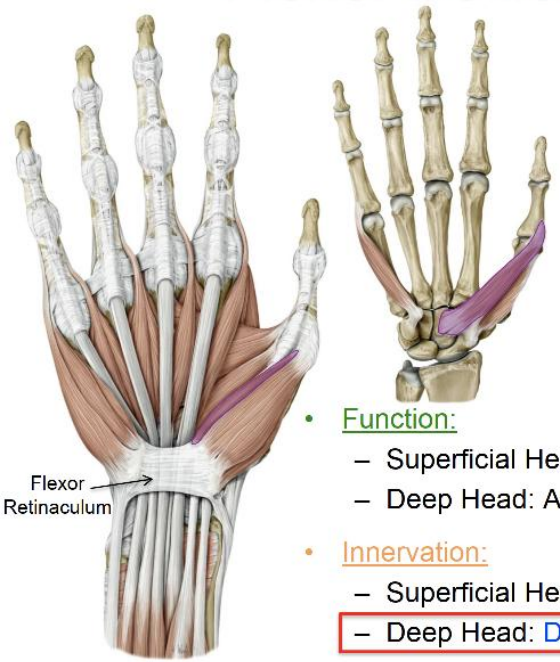


Insertion:
Lateral base of proximal phalanx
Lateral head of metacarpal

Origin:
Flexor retinaculum
Scaphoid & Trapezium

- Function:
  - Abduction of CMC joint
- Innervation:
  - Recurrent br. of median nerve

Flexor Pollicis Brevis

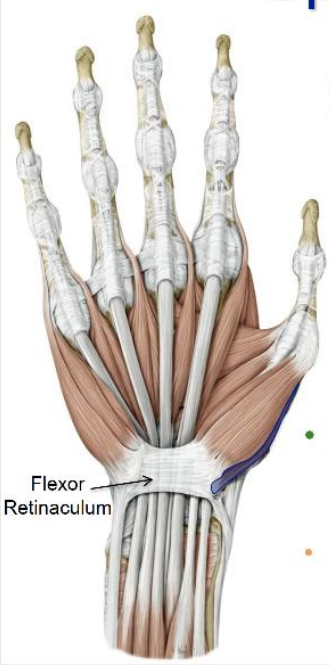


Insertion:
Lateral base of proximal phalanx
Lateral head of metacarpal

Origin:
Flexor retinaculum
Capitate & Trapezium

- Function:
  - Superficial Head: Flexes CMC joint
  - Deep Head: Adducts CMC joint
- Innervation:
  - Superficial Head: Recurrent br. of median nerve
  - Deep Head: Deep br. of ulnar nerve

Opponens Pollicis







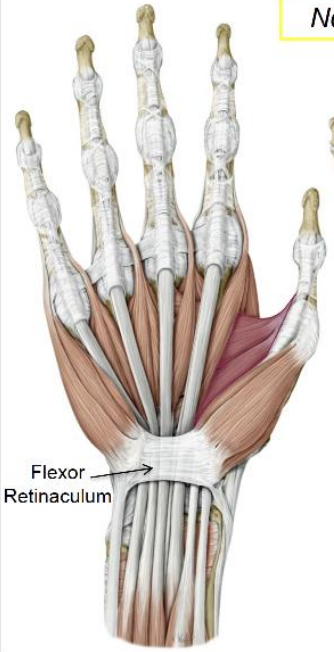
Insertion:
Lateral aspect of 1st metacarpal

Origin:
Trapezium
Flexor retinaculum

- Function:
  - Opposition of CMC joint
- Innervation:
  - Recurrent br. of median nerve

Adductor Pollicis

Not part of thenar compartment







Origin:
Transverse head:
Body MC 3
Oblique head:
Base of MC 2-3
Capitate

Insertion:
Medial side of base of proximal phalanx and head of MC

- Function:
  - Adduction of CMC joint
  - Flexion of MCP joint
- Innervation:
  - Deep br. of ulnar nerve

Flexor Retinaculum

a. Fibrous band extending between the outer carpal bones that forms the roof of the ***carpal tunnel.***
b. Passing superficial to the flexor retinaculum is the ***ulnar nerve & artery***, a ***palmar branch*** of the ***median nerve***, and the ***palmaris longus tendon*** (if present).
c. Passing deep to the flexor retinaculum (within the carpal tunnel) are:
   i. ***Flexor digitorum superficialis*** (4 tendons)
   ii. ***Flexor digitorum profundus*** (4 tendons)
   iii. ***Flexor pollicis longus***
   iv. ***Median nerve***

B. Intrinsic Muscles of the Hand

1. Thenar Muscles:

Muscle	Origin	Insertion	Innervation	Main Action
Abductor pollicis brevis	Flexor retinaculum, scaphoid & trapezium	Lateral side of base of proximal phalanx of thumb & head of MC	**Recurrent branch of the median n.**	Abducts thumb
Flexor pollicis brevis Superficial head Deep head	Flexor retinaculum, capitate & trapezium	Lateral side of base of proximal phalanx of thumb & head of MC	*Superficial head:* **Recurrent branch of the median n.** *Deep head:* **Deep branch of ulnar n.**	*Superficial head:* Flexes thumb *Deep head:* Adducts thumb
Opponens pollicis	Flexor retinaculum, trapezium	Lateral side of 1st metacarpal	**Recurrent branch of the median n.**	Opposition of CMC joint
Adductor pollicis Oblique head Transverse head *(not true thenar muscle)*	*Oblique head:* base of metacarpals 2M, capitate, & adjacent carpals *Transverse head:* body of metacarpal 3	Medial side of base of proximal phalanx of 1st digit	**Deep branch of ulnar n.**	Adducts thumb

Clinical Note: Functional deficits in ***adductor pollicis*** are tested by observing the ability of the patient to squeeze & retain a piece of paper between digits 1 & 2 when gently try to tug it away

C. Extrinsic Muscles of the Hand (Review)

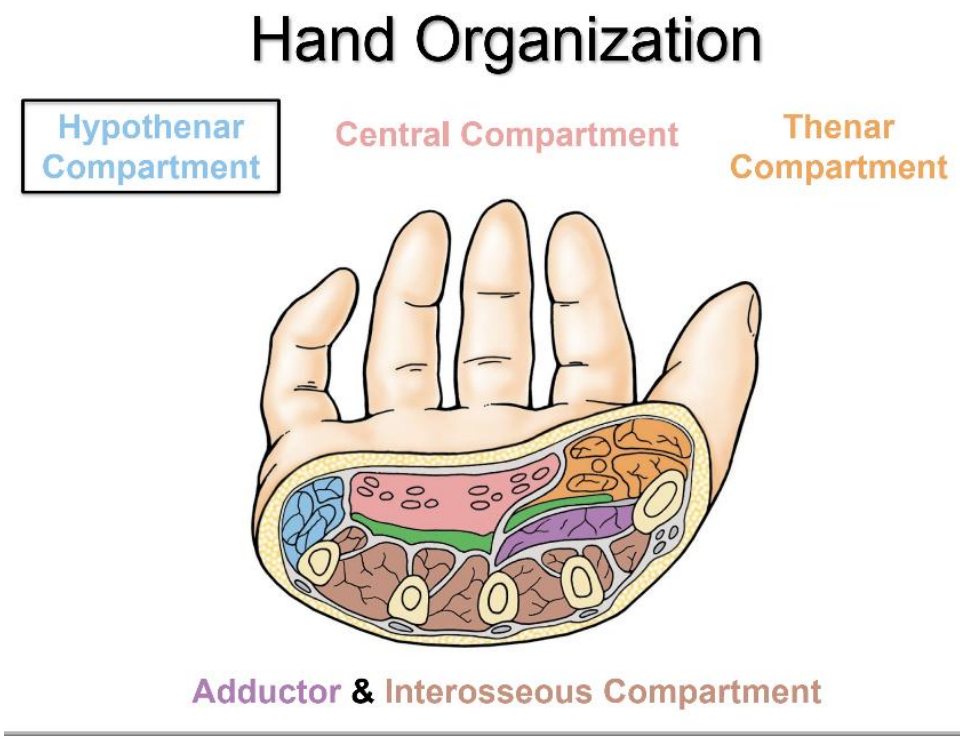
a. **Flexion at Wrist:** Flexor Carpi Ulnaris, Palmaris Longus, & Flexor Carpi Radialis

b. **Flexion of Digits:**

i. **Flexor pollicis longus** – tendon runs between 2 sesamoid bones and inserts on the base of distal phalanx of the 1st digit
ii. **Flexor digitorum profundus (FDP)** – tendons (4) insert on base of distal phalanges of digits 2-5. It is the only muscle that can flex the DIP joint.
iii. **Flexor digitorum superficialis (FDS)** – tendons (4) split & insert on middle phalanges of digits 2-5. FDP tendons pass through/deep to the FDS tendons to insert on distal phalanx base. FDS can flex all joints that it crosses but has its greatest strength at the wrist joint.

c. **Fibrous digital sheaths** provide osseofibrous canals that closely bind tendons to the phalanges. They function as a series of pulleys and allow tendons to glide through synovial sheaths containing synovial fluid, which lubricates them & decreases friction.

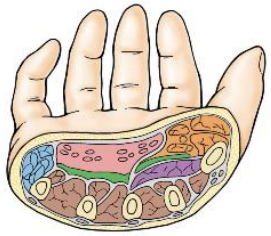
Clinical Note: Injuries to the fingers can cause infection of the digital synovial sheaths. Inflammation of the tendon & synovial sheath or **tenosynovitis** causes the finger to swell and become painful to move. Since the digits have separate synovial sheaths, infection is usually confined to the infected digit, but may spread through connections to the common flexor sheath and potentially to the anterior forearm



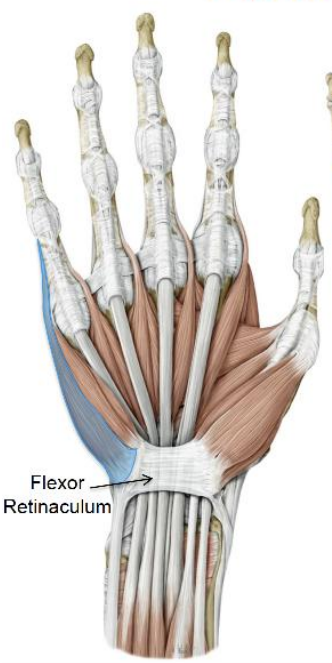
Hypothenar Compartment



- Act upon the little finger (digit 5)
- Three muscles
- Make up hypothenar eminence
- Innervated by: **Deep branch of ulnar nerve**



Abductor Digiti Minimi





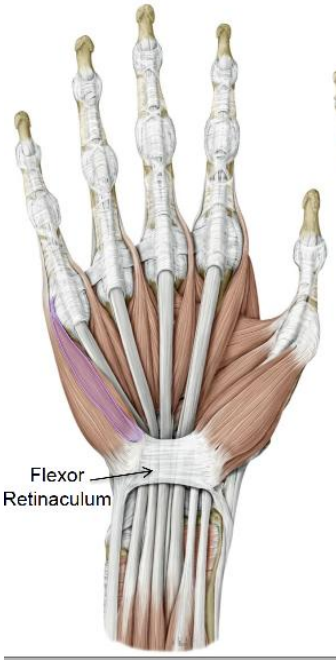


Insertion:
Medial base of proximal phalanx

Origin:
Pisiform

- Function:
  - Abduction of 5th digit
- Innervation:
  - Deep br. of ulnar nerve

Flexor Digiti Minimi Brevis





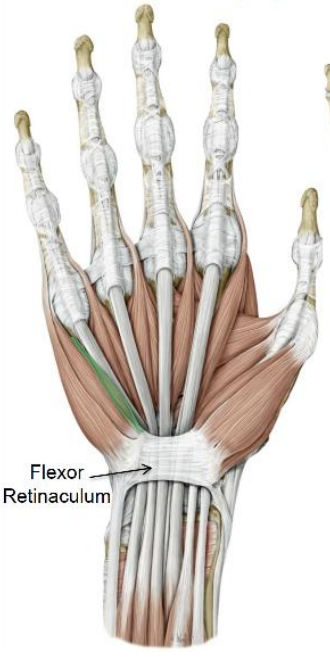


Insertion:
Medial base of proximal phalanx

Origin:
Flexor retinaculum
Hook of hamate

- Function:
  - Flexion of 5th digit
- Innervation:
  - Deep br. of ulnar nerve

Opponens Digiti Minimi







Insertion:
Medial border of body of 5th metacarpal

Origin:
Hook of hamate
Flexor retinaculum

- Function:
  - Opposition of the 5th digit
- Innervation:
  - Deep br. of ulnar nerve

2. Hypothenar Muscles:

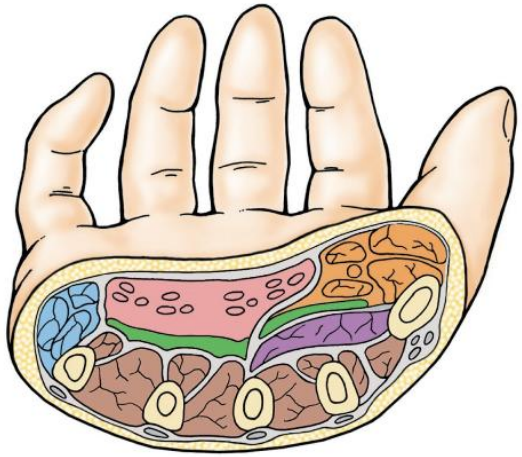
Muscle	Origin	Insertion	Innervation	Main Action
Abductor digiti minimi	Pisiform	Medial side of base of proximal phalanx of 5th digit	**Deep branch of ulnar n.**	Abducts 5th digit
Flexor digiti minimi brevis	Flexor retinaculum & hook of hamate	Medial side of base of proximal phalanx of 5th digit	**Deep branch of ulnar n.**	Flexes 5th digit
Opponens digiti minimi	Flexor retinaculum & hook of hamate	Medial border of 5th metacarpal	**Deep branch of ulnar nerve**	Opposition & rotation of 5th digit

Hand Organization

Hypothenar Compartment

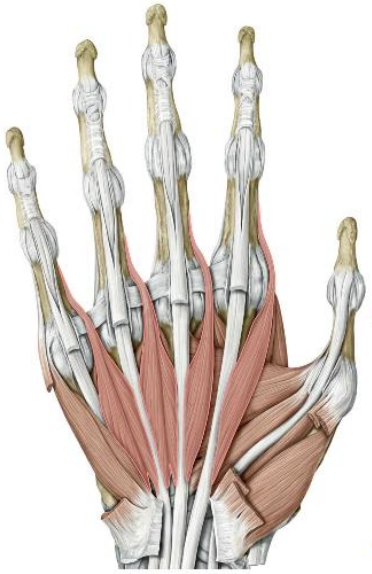
Central Compartment

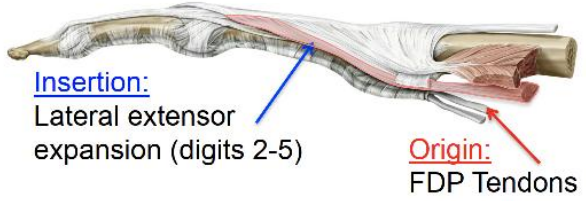
Thenar Compartment



Adductor & Interosseous Compartment

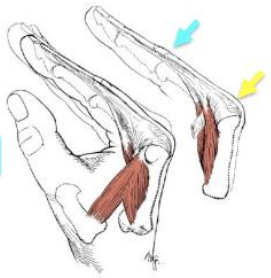
Lumbricals





Insertion:
Lateral extensor expansion (digits 2-5)

Origin:
FDP Tendons



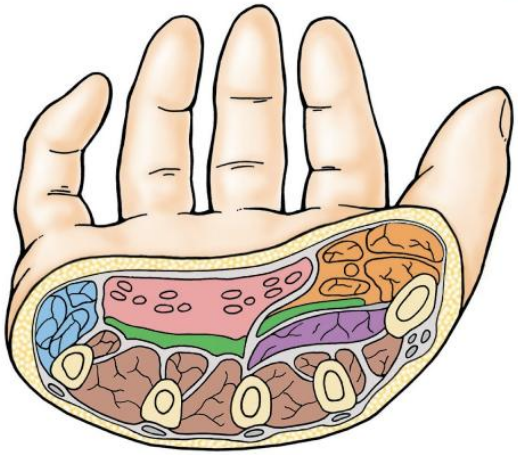
- Function:
  - Flex the MCP joint
  - Extend the IP joints
- Innervation:
  - Medial Two: Deep br. of ulnar nerve
  - Lateral Two: Median nerve

Hand Organization

Hypothenar Compartment

Central Compartment

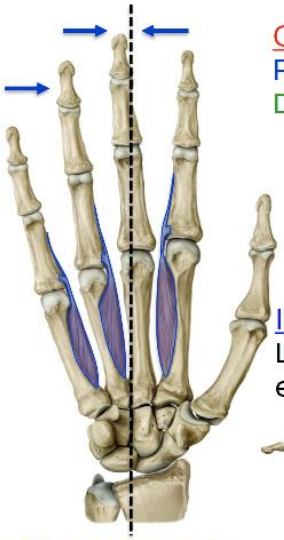
Thenar Compartment



Adductor & Interosseous Compartment

Interossei

Origin:
PAD: Midline MC 2, 4, 5
DAB: Adjacent two MC



Insertion:
Lateral extensor expansion (digits 2-5)





Palmar Interossei (3)
Adduction (P-A-D)

Dorsal Interossei (4)
Abduction (D-A-B)

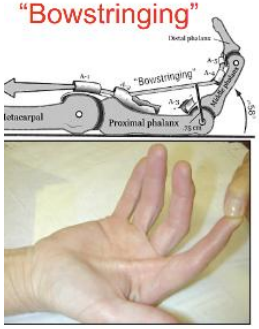
Palmar View

Digital Sheaths

Fibrous Digital Sheaths

Osseofibrous canals that bind tendons to the phalanges

"Bowstringing"

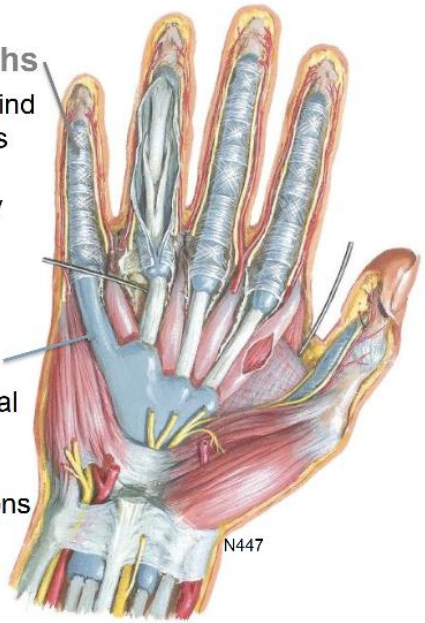


Function: Act as a pulley system for force of long tendons

Synovial Sheaths

Bursae containing synovial fluid

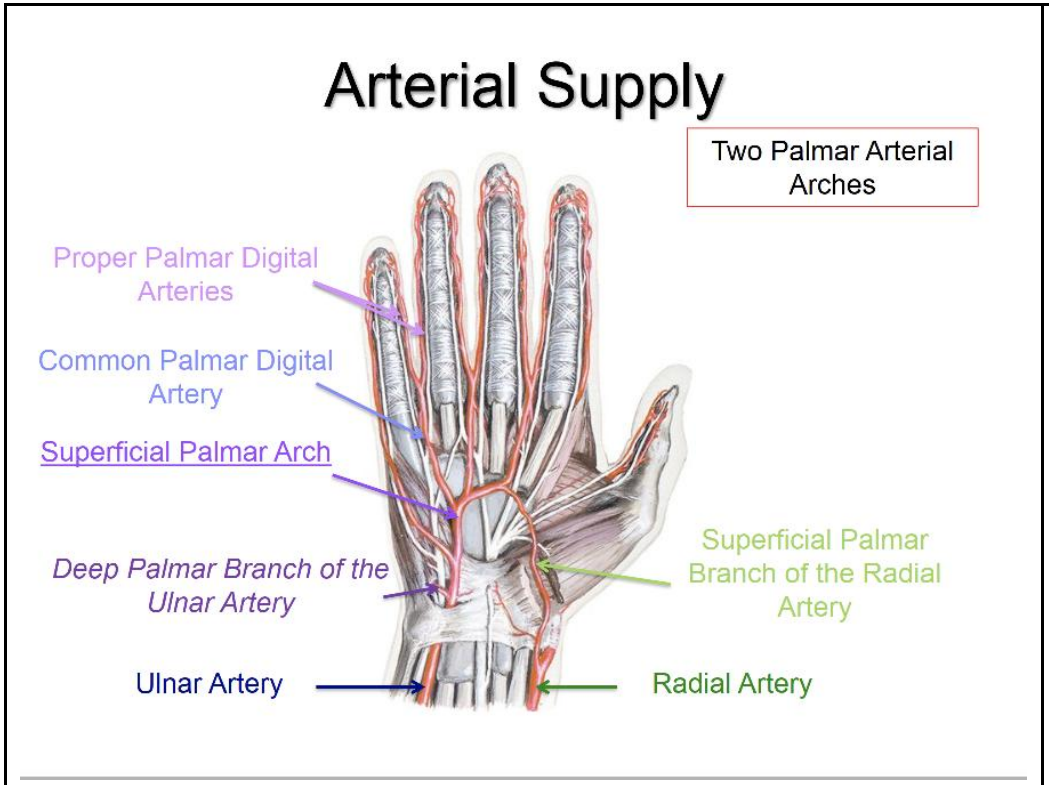
Function: Lubricates tendons and decreases friction

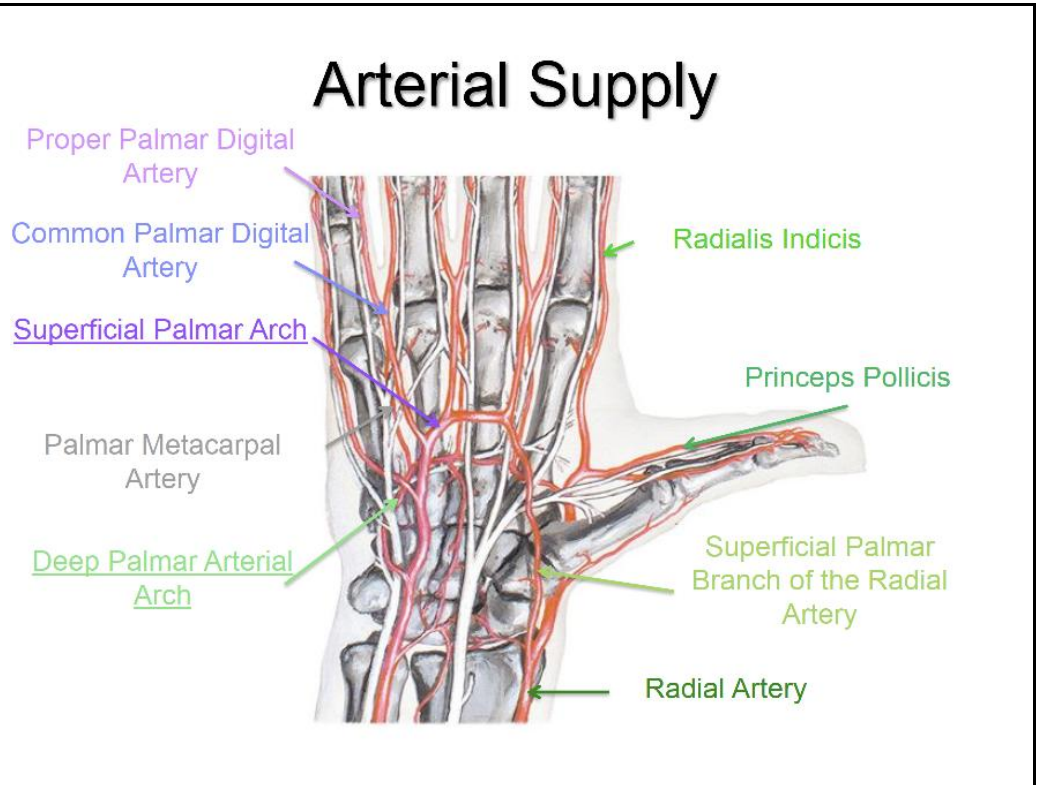


Clinical Note: Functional deficits of ***dorsal interosseous muscles*** are tested by observing the ability of the patient to spread digits 2-3 apart against resistance by the examiner. To test ***palmar interosseous muscles***, the patient is asked to hold a sheet of paper between adjacent fingers and prevent the paper from being pulled away by the examiner.

1. Short Hand Muscles:

Muscle	Origin	Insertion	Innervation	Main Action
Palmar interosseous muscles (3)	Palmar surfaces of metacarpals 2, 4, & 5	Base of proximal phalanges & extensor expansions of digits 2, 4, & 5	**Deep branch of the ulnar nerve**	**Adduct** digits 2, 4, & 5 to the midline (3rd digit) = ***PAD*** Assists lumbricals in flexing MP joints & extending IP joints
Dorsal interosseous muscles (4)	Adjacent sides of two metacarpals	Base of proximal phalanges & extensor expansions of digits 2- 4	**Deep branch of the ulnar nerve**	**Abduct** digits 2-4 from the midline = ***DAB*** Assists lumbricals in flexing MP joints & extending IP joints
Lumbricals 4 "worm-like" muscles	1-2: Lateral 2 FDP tendons 3-4: Medial 3 FDP tendons	Lateral sides of extensor expansions of digits 2-5	*Digits 2 & 3:* **Median nerve** *Digits 4 & 5:* **Deep branch of the ulnar n.**	Flex the MP joint & extend IP joints of digits 2-5 (*Z-movement*)





D. Blood Supply to the Hand

1. Ulnar Artery

a. Courses superficial to the flexor retinaculum between the pisiform & hook of the hamate via the ulnar canal (***Guyon's canal***) to enter the hand.
b. Gives rise to the ***deep branch of the ulnar artery*** that dives through the hypothenar compartment and contributes to the ***deep palmar arch***
c. Ulnar artery continues & contributes to the ***superficial palmar arch*** (main contributor). Distally the superficial palmar arch is protected by passing deep to the palmar aponeurosis.

2. Radial Artery

a. Usually sends a small superficial branch to contribute to the superficial palmar arch before passing dorsal to the palm in the "*anatomical snuffbox*". It enters the palm by diving between the heads of the 1st dorsal interosseous muscle.
b. Gives off ***princeps pollicis artery*** (to thumb) & ***radialis indicis artery*** (to lateral side of 2nd digit).
c. Radial artery ends by anastomosing with the deep branch of the ulnar artery to form the ***deep palmar arch*** (main contributor).

3. Superficial Palmar (Arterial) Arch

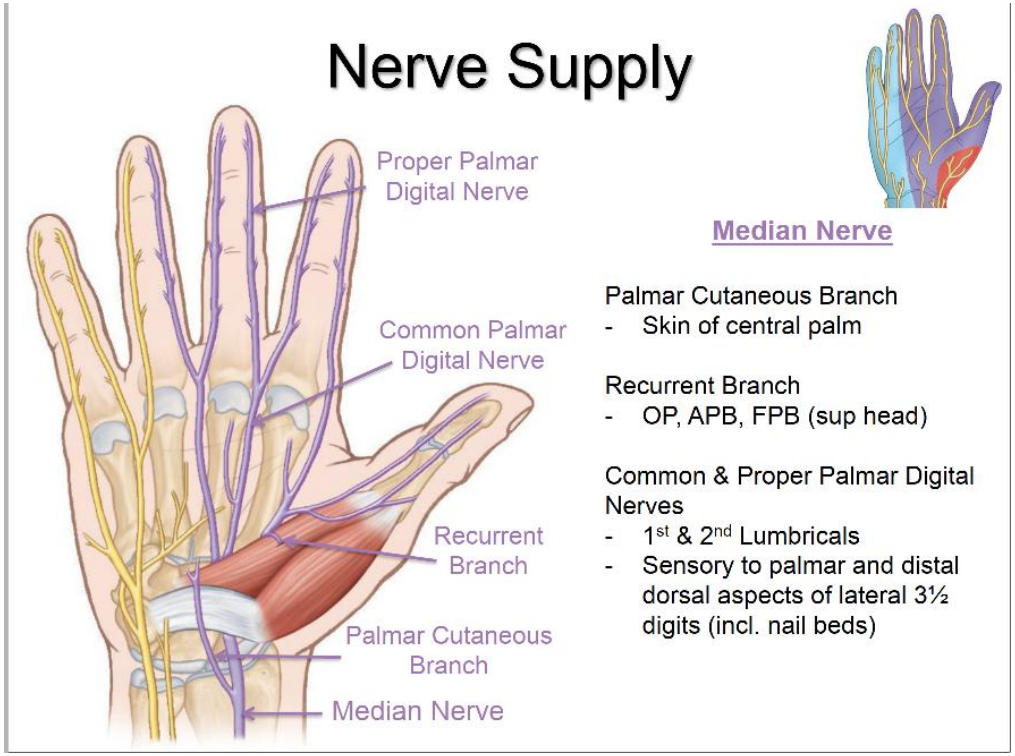
a. Located immediately deep to the palmar aponeurosis.
b. Mainly a continuation of the ulnar artery, although it is usually completed by a superficial branch of the radial artery.
c. Gives rise to **common palmar digital arteries** that branch into **proper palmar digital arteries**

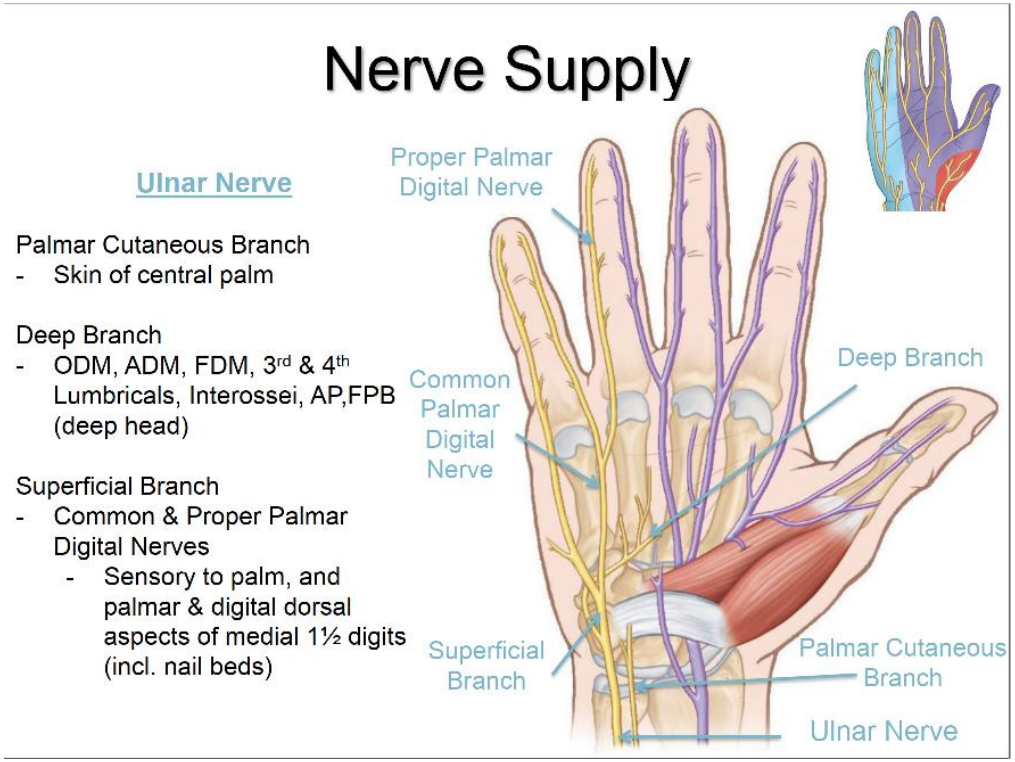
4. Deep Palmar (Arterial) Arch

a. Lies across the metacarpals just distal to their bases.
b. Mainly a continuation of the deep branch of the radial artery, but it may be completed by a deep branch of the ulnar artery.

c. Gives rise to **3 palmar metacarpal arteries.**

Nerve Supply of the Hand





E. Nerve Supply to the Hand

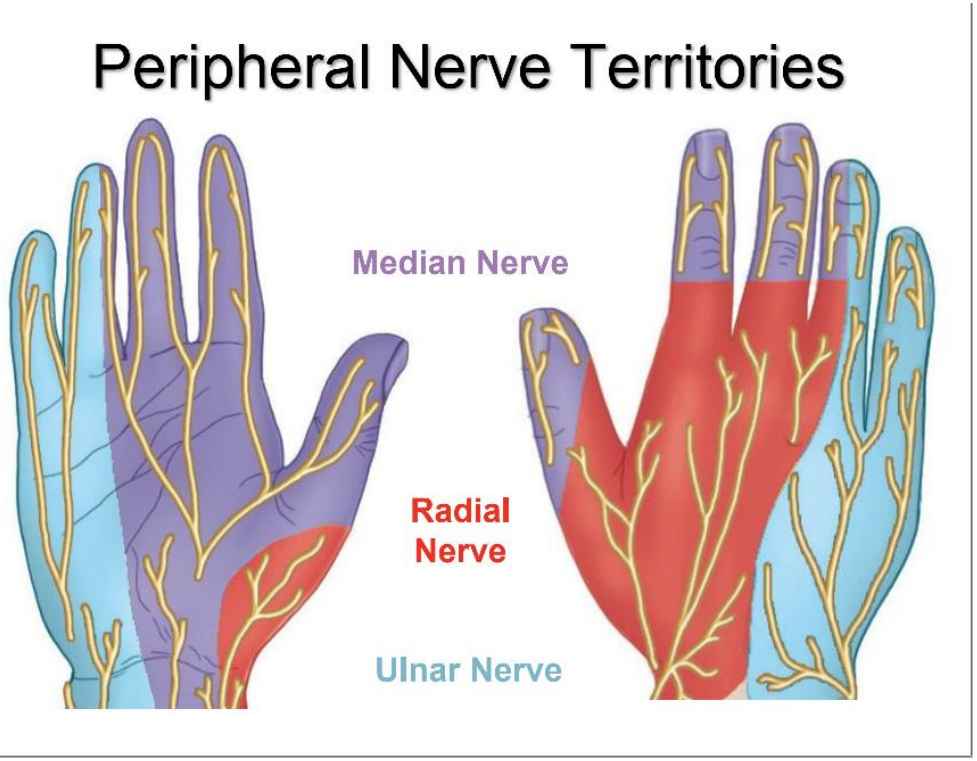
1. Median Nerve

a. **Palmar cutaneous nerve** – branch proximal to **carpal tunnel** to innervate skin of central palm.
b. Enters the hand through the ***carpal tunnel*** deep to the flexor retinaculum.
c. **Recurrent branch of the median nerve** – runs superficial course distal to the carpal tunnel
   i. Innervates ***thenar muscles:*** abductor pollicis brevis, opponens pollicis, & flexor pollicis brevis.
   ii. At risk with a superficial cut to the lateral side of the palm. It must be repaired or the thenar muscles will atrophy
d. ***Median nerve branches*** innervate the 1st & 2nd lumbricals
e. Sends sensory fibers to the skin on the lateral palmar surface (via ***common palmar digital branches***) and the palmar aspect of the lateral 3½ digits and distal dorsal halves (and nail beds) of these digits (via ***proper palmar digital branches***).

2. Ulnar Nerve

a. Proximal to the wrist:
   i. Palmar cutaneous branch passes superficial to the flexor retinaculum & palmar aponeurosis to supply the skin on the medial palm.
   ii. Dorsal cutaneous branch supplies skin on medial half of the dorsum of the hand and proximal portions of the dorsal medial 1½ digits.
b. Passes through the ***ulnar*** or ***Guyon's canal*** (fibro-osseous arch stretching between the pisiform & hook of the hamate). Enters hand lateral to the pisiform bone and divides into superficial & deep branches.
c. Superficial branch of the ulnar nerve supplies anterior skin of the medial 1½ digits (via ***common palmar digital branches*** and ***proper palmar digital branches***).
d. Deep branch of ulnar nerve innervates the hypothenar muscles, medial 2 lumbricals, adductor pollicis, all the interosseous muscles, and deep head of flexor pollicis brevis. It supplies sensation to the metacarpal bones and

proprioception to the joints.



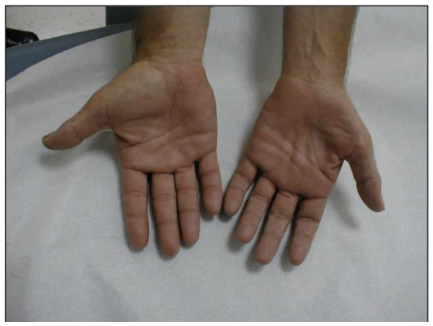
Nerve Lesions

Median Nerve

Signs & Symptoms:

Fill in

- can still flex/extend, abduct
- cannot do opposition
- Signs: atrophy of thenar eminence





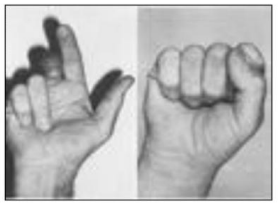
Sites at Risk of Injury

Nerve Lesions

Median Nerve

Signs & Symptoms:

Hand of benedictmask pt to make fist and instead of fist, they end up in this position



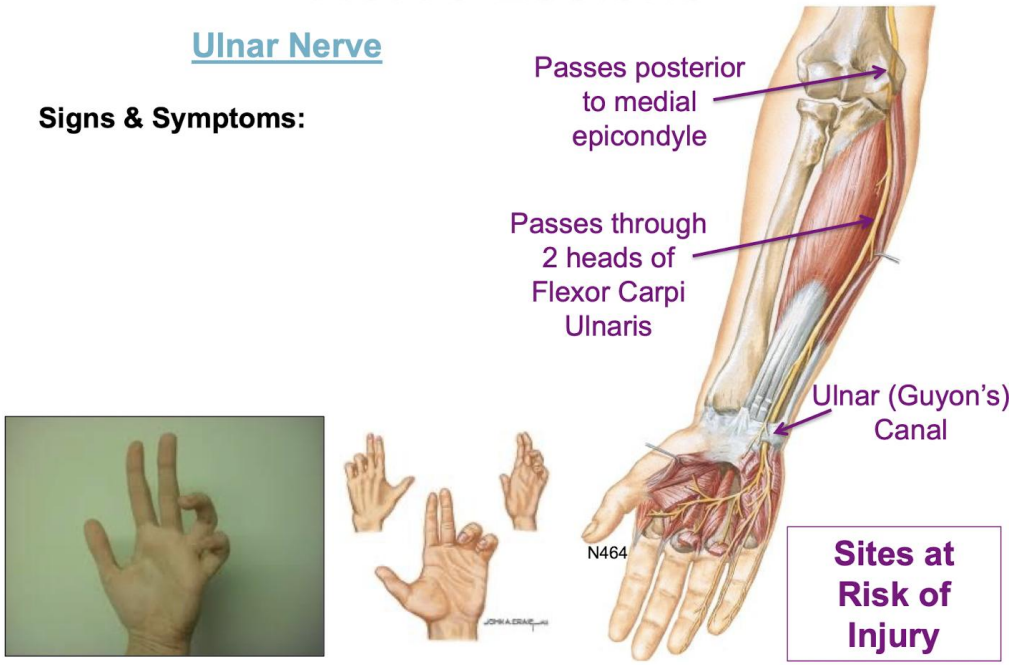




Sites at Risk of Injury

Clinical Note: Lesions of the **median nerve** usually occur in the forearm or wrist, with the most common site being the carpal tunnel. Lacerations of the wrist often cause median nerve injury because the nerve is relatively close to the surface. This results in paralysis and wasting of the thenar muscles and the first two lumbricals. Opposition of the thumb is not possible, fine control movements of the 2nd & 3rd digits are impaired, and sensation is lost over the thumb and adjacent 2½ fingers.

- Median nerve lesions in the elbow region result in loss of flexion of the PIP & DIP joints (FDS & FDP) and flexion of the MP joints (lumbricals) of the 2nd & 3rd digits. This injury results in a "hand of benediction" when trying to make a fist. Relatively minor lacerations of the thenar eminence can sever the recurrent branch of the median nerve as it lies subcutaneously, which would paralyze the thenar muscles.



Clinical Note: Lesions of the ulnar nerve usually occur in one of 4 places:

- Posterior to the medial epicondyle of the humerus (most common)
- In the cubital fossa between the 2 heads of flexor carpi ulnaris
- At the wrist
- In the hand.

Lesions at any of these sites result in **extensive motor & sensory loss** to the hand. With a proximal ulnar nerve injury, wrist flexion is impaired, and the hand is drawn to the lateral side by FCR (in the absence of FCU function).

- The patient would have difficulty making a fist because, in the absence of opposition, the MP joints become hyperextended, and he/she cannot flex digits 4-5 at the DIP joints.
- The person also cannot extend the IP joints when trying to straighten the fingers, creating the characteristic ulnar "**claw hand**" due to the atrophy of the interosseous muscles of the hand. The claw is produced by the unopposed action of the extensors and FDP.
- Compression of the ulnar nerve at the wrist in Guyon's canal can cause ***ulnar canal syndrome,*** resulting in reduced sensation in the medial 1½ fingers and weakness of the intrinsic hand muscles.
    - Clawing of the 4th & 5th fingers may occur, but in contrast to a proximal ulnar nerve injury, their ability to flex is unaffected and there is no radial deviation of the hand.

Clinical Note: Upper brachial plexus injuries (Erb-Duchenne Palsy) affect C5-C6 and usually occur due to an excessive increase in the angle between the neck & shoulder (e.g., being thrown from a motorcycle, excessive stretching of a newborn's neck during delivery).

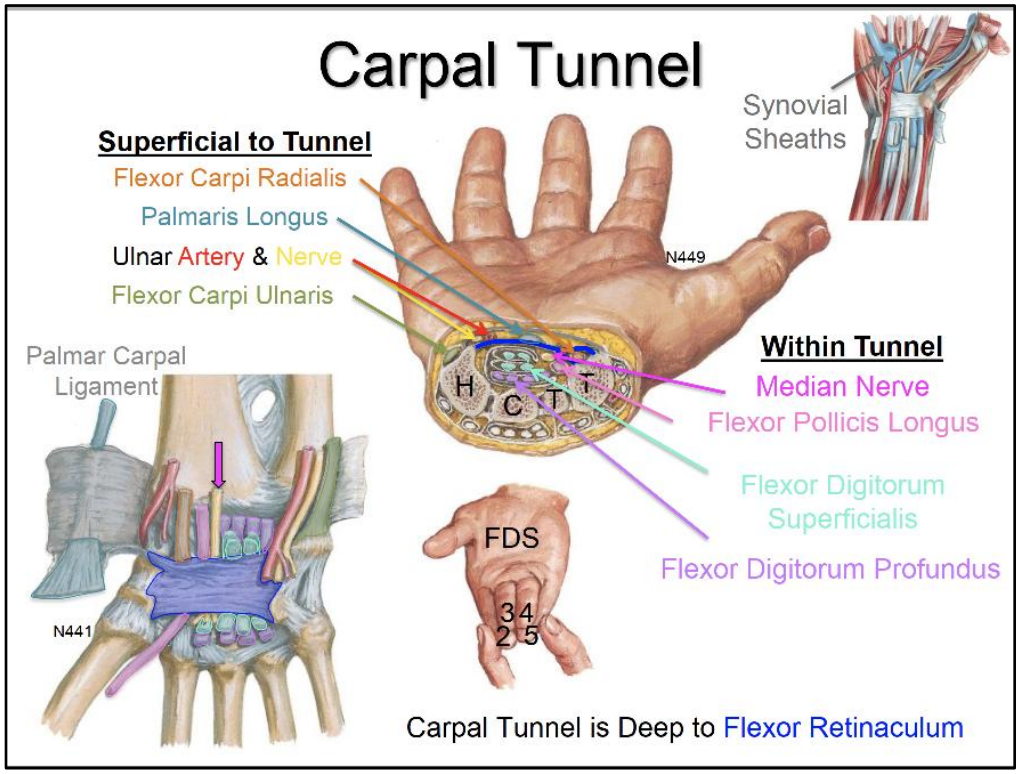
- This stretches or tears the superior trunk or roots of the brachial plexus from the spinal cord.
- **Symptoms** include an upper limb with an adducted shoulder, medially rotated arm, and extended elbow ("waiter's tip position") due to paralysis of the shoulder & arm muscles supplied by C5-C6.
    - There is also a loss of sensation on the lateral aspect of the upper limb.

Clinical Note: Lower brachial plexus injuries (Klumpke Paralysis) are less common but affect C8-T1 and usually occur when the upper limb is suddenly pulled superiorly (e.g., a baby's upper limb being pulled excessively during delivery, when a person grasps something to break a fall).

- This stretches or tears the inferior trunk or roots of the brachial plexus from the spinal cord.
- This would affect the ulnar nerve, resulting in the intrinsic hand muscles being affected causing a claw hand.

Carpal Tunnel Syndrome

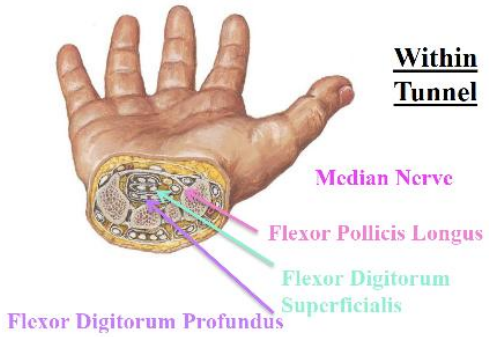
Carpal Tunnel



Carpal Tunnel Syndrome

- The bread and butter peripheral nerve diagnosis
- Compression of the median nerve at the wrist (in the "carpal tunnel")

- **Floor: carpal bones**
- **Roof: transverse carpal ligament "flexor retinaculum"**
- **Contents: tendons of FPL, FDP, FDS, median nerve**



Carpal Tunnel Syndrome

- Causes
  - Extrinsic (reduced space)
    - Fracture
    - Flexor tenosynovitis (finger flexor tendons share space in the carpal tunnel)
    - Ganglia
    - Inflammation (rheumatologic conditions)
    - Pregnancy (edema)
    - Acromegaly
  - Intrinsic (increased susceptibility)
    - Diabetes
    - Amyloidosis
    - Alcohol
- And, very commonly…idiopathic

Carpal Tunnel - osseofibrous tunnel created by the hamate, capitate, trapezoid, and trapezium and the flexor retinaculum. The flexor retinaculum passes from the scaphoid and trapezium laterally and the pisiform and hook of the hamate medially. The following structures pass through the carpal tunnel: flexor pollicis longus tendon, all four flexor digitorum superficialis tendons, all four flexor digitorum profundus tendons, and the median nerve.

Clinical Note: **Carpal tunnel syndrome** results from any lesion that significantly reduces the size of the carpal tunnel.

- Most commonly caused by inflammation of the synovial sheaths (synovial sheaths surround tendons passing through osseofibrous tunnels to reduce friction).
- Fluid retention, infection and excessive overuse may cause swelling of the tendons or their synovial sheaths.
- Median nerve is the most sensitive structure in the tunnel and may become compressed. Compression of the median nerve may result in hypoesthesia of the lateral palm and lateral 3.5 digits as well as weakness to the intrinsic muscles of the thumb.
- A branch of the median nerve, **palmar cutaneous branch**, does not pass through the carpal tunnel and thus its area of cutaneous innervation, a small patch of the mid-palm, will not be affected by carpal tunnel syndrome.

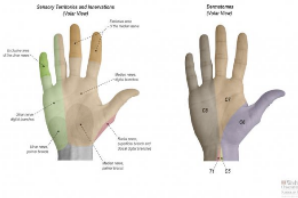
Clinical Presentation

- Numbness and tingling involving the territory of the median nerve
  - Often predominantly at night
  - Flexed/extended wrist = worsening (driving, holding phone)
- Potentially thumb weakness (APB)
- Median function above the carpal tunnel is normal
  - Median innervated forearm muscles!
- Maybe tinel and phalen's signs



Differential Diagnosis

- What might this get confused with?
  - Main item on differential is C6 radiculopathy
  - Others may include
    - Tendinopathy
    - CMC OA
- EMG/NCS very helpful
  - Distinguish muscle/bone from nerve problem
  - Distinguish CTS from C6
  - Establish severity of nerve damage



Management

(from Approach to Management of ALL THINGS MSK)

- Education
  - Common problem, education about hand position and reversibility if discovered early
- Non-Pharm
  - Rigid wrist hand orthosis worn overnight only
  - Avoid aggravating positions, vibratory tools, etc
- Pharm
  - Corticosteroid injection of carpal tunnel
- Intervention/surgery
  - Carpal tunnel release

III. Testing Spinal Cord Levels

A. Testing Myotomes:

1. **C5** – Abduction of the arm and flexion of the forearm
2. **C6** – Flexion of the forearm and extension of the wrist
3. **C7** – Extension of the forearm, flexion of the wrist, and extension of the digits
4. **C8** – Flexion of the digits
5. **T1** – Abduction and adduction of the digits

B. Testing Dermatomes:

1. **C5** – Lateral elbow
2. **C6** – Anterior distal 1st digit
3. **C7** – Anterior distal 3rd digit
4. **C8** – Anterior distal 5th digit
5. **T1** – Medial elbow